PTYS578 Assignment 10

The Mysterious Transit Depths of 55 Cancri e: JWST Observations to Constrain the Presence of Atmospheric Silicates

55 Cancri e represents the most common type of exoplanet, one for which there is no counterpart in the Solar System: the Super Earth. Establishing the presence and constraining the composition of atmospheres on Super Earths is essential to advance our understanding of planetary formation, evolution and composition, and to inform future studies of rocky planets and habitability. However, 55 Cancri e has demonstrated a puzzling property that cannot be explained by our current data: a significant variation in secondary eclipse transit depths over the course of multiple observations. We propose to perform simultaneous JWST MIRI and NIRSpec observations to constrain the presence of silicates in 55 Cancri e's atmosphere. Observations of silicates would support a possible cloud feedback mechanism that would explain the discrepancies in transit depths. Even if silicates are not observed, this program will still be an opportunity to robustly confirm and characterize the first ever rocky exoplanet atmosphere. The proposed observations will establish 55 Cancri as a gold-standard benchmark that will enable interpretation of future observations of Super Earths and will give us entirely new insight into planetary formation and climate as a field.

- **Scientific Justification**

55 Cancri e is a window into the nature of Super Earth exoplanets

55 Cancri e represents the most common type of exoplanet, one for which there is no counterpart in the Solar System: the Super Earth. Given its proximity to its host star and its resulting 18-hour orbital period, it is the perfect target for observations over the course of multiple orbits [1]. This planet has been subject to various prior observations that have heavily implied the existence of an atmosphere; however, there is not yet enough data to definitively confirm that or to determine the alleged atmosphere's precise composition [2]. **Establishing the presence and constraining the composition of an atmosphere on a Super Earth is essential to advance our understanding of planetary formation, evolution, and composition, and 55 Cancri e is an ideal candidate in the pursuit of this goal.**

However, 55 Cancri e has demonstrated a puzzling property that cannot be explained by our current data: a significant variation in secondary eclipse transit depths over the course of multiple observations [3]. Understanding this variation is critical not only to our understanding of 55 Cancri e individually, but also Super Earths as a population, the characterization which has only just become possible with the advent of JWST.

We propose to perform simultaneous JWST MIRI and NIRSpec observations to definitively constrain a. the existence of an atmosphere on 55 Cancri e, and b. the presence of silicates in its atmosphere which would drive a feedback mechanism that would explain the variations in secondary eclipse transit depth. JWST is the only instrument with the necessary wavelength range to capture the spectral signature of the silicates that would enable this mechanism, and this work is timely in its response to the most recent Astronomy and Planetary Decadal Surveys' calls to identify and characterize the atmospheres of rocky worlds as we prepare for upcoming missions such as the Habitable Worlds Observatory. Ultimately, these observations will answer a persistent question specific to 55 Cancri e and establish it as a benchmark for interpretation of future observations of Super Earths generally.

Silicates can explain variations in 55 Cancri e's secondary eclipse transit depths

Understanding the nature of 55 Cancri e will open the door to future studies of Super Earth exoplanets. However, previous observations have shown significant variations in 55 Cancri e's secondary eclipse transit depth over the course of multiple orbits which current data cannot explain. A proposed solution to this puzzle is a feedback mechanism driven by silicates, wherein the radiation from the parent star evaporates silicates from the planet's proposed magma ocean and these silicates form clouds in the cooler upper atmosphere of the planet. These clouds then obstruct incoming radiation, causing the planet to cool and preventing further silicates from evaporating, causing the clouds to eventually dissipate [4]. Thus, when 55 Cancri e passes behind its star during a "cool" phase (when there are clouds in the atmosphere) it produces a significantly shallower secondary eclipse transit depth than when it passes behind the star during its "hot" phase (when there are no clouds in the atmosphere). Definitively identifying silicates in the atmosphere of 55 Cancri e would strongly support this hypothesis.

To this end, we will perform simultaneous observations with JWST MIRI and NIRSpec covering the full wavelength range in which silicates appear. These observations will not only provide us with the data necessary to confirm or deny the silicate feedback hypothesis to explain the transit depth variations, but this full wavelength range will also allow observation of other atmospheric components, allowing for the much-needed definitive confirmation of the existence of an atmosphere on 55 Cancri e. This would be the first-ever definitive confirmation of a rocky exoplanet atmosphere.

<u>This Proposal: Constrain the atmospheric composition of 55 Cancri e through JWST transit spectroscopy</u>

This work will acquire transit spectroscopy of the Super Earth exoplanet 55 Cancri e. JWST is the only facility with the sensitivity necessary to observe a small, rocky world and the instruments that cover the full wavelength range of interest to this study.

We propose to take simultaneous JWST MIRI and NIRSpec transit observations to a) unequivocally confirm the presence and b) identify the components of the atmosphere of 55 Cancri e. These instruments combined cover the entire wavelength range in which the spectral signatures of silicates appear. Figure 1 demonstrates how silicates in the atmosphere would drive a cloud feedback mechanism that would explain the puzzling variations in secondary eclipse transit depth that have been observed.

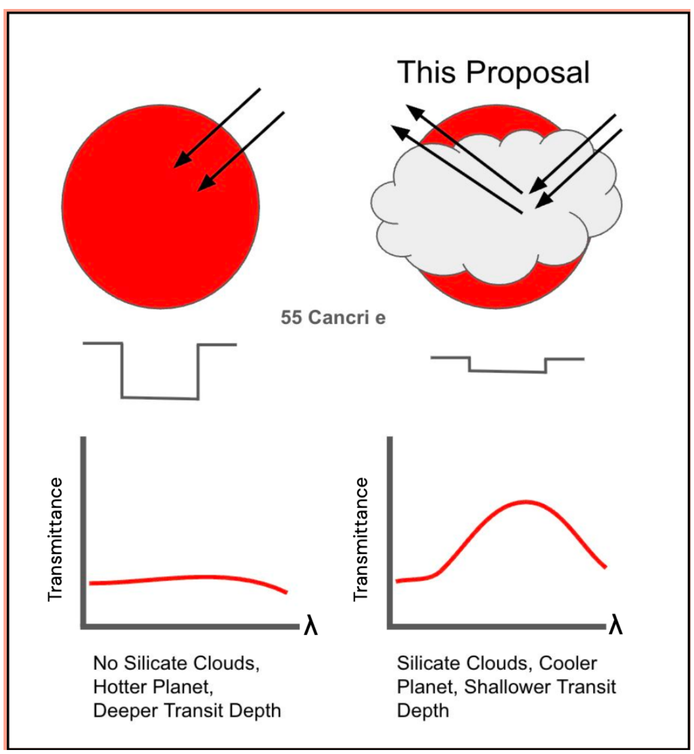

Figure 1: ***The confirmation of silicate spectral features in the atmosphere of hot super-earth 55 Cancri e would support the existence of a cloud feedback mechanism that can explain the observed variations in secondary eclipse occultation depth.*** *On the left is a cloud-free planet with a hotter surface due to a lack of clouds blocking incoming radiation, which would result in a deeper secondary eclipse depth. On the right is a planet with silicate clouds in its atmosphere, which would result in a cooler planet and therefore a shallower secondary eclipse transit depth.* *<u>We propose to perform simultaneous JWST MIRI and NIRSpec observations to probe the wavelengths at which such silicate features would appear,</u>* *which will allow us to constrain 55 Cancri e's atmospheric composition, explain its puzzling discrepancies in transit depth, and give us insight into an as-yet unexplored mechanism for driving climate on close-in rocky exoplanets.*

The transit spectra of 55 Cancri e we obtain from this work will confirm or deny the presence of silicates in the planet's atmosphere. **The detection of silicates would support a cloud feedback hypothesis that would solve the persistent problem of variations in secondary eclipse transit depths that had been seen in previous observations of 55 Cancri e.**

<u>55 Cancri e ushers in a new era of understanding rocky exoplanets</u>

The observing campaign we have proposed will allow us to test the hypothesis that a feedback mechanism driven by silicate clouds is behind the varying secondary eclipse transit depths seen in previous observations of 55 Cancri e. **Moreover, these observations will provide the necessary data to robustly confirm a rocky exoplanet atmosphere for the first time, marking a significant milestone in exoplanet science and enabling interpretation of future observations of such planets.**

This work is only possible with the advent of JWST, which is giving us the capability to observe small, rocky, planets for the first time. One of JWST's primary goals is to perform transit spectroscopy on these never-before-studied targets. Additionally, there is significant emphasis on the importance and timeliness of constraining a rocky planet's atmosphere in both the planetary science and astronomy decadal surveys [5, 6]. As such, the time is ripe for this work to be carried out, and doing so is essential for expanding our understanding of the entire planetary population, as well as providing specific insight into the formation and evolution of planetary systems. **Constraining the presence and composition of 55 Cancri e's atmosphere and identifying a key process driving its climate are the first steps into a new era of rocky exoplanet science.**

References

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